Varying index coefficient modeling for environmental risk assessment

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1 Proposal description

There is a growing body of evidence showing that exposure to environmental risks such as polluted air, contaminated food and water, poor sanitation, indoor smoke, infectious disease vectors, hazardous chemicals, etc. may adversely impact child development, although research on this remains extremely limited. Due to their size, behavior and physiology, children are more vulnerable than adults to environmental hazards. Moreover, they are more heavily exposed to environmental contaminants in proportion to their weight, and as a result they may suffer long-term effects from early exposure. Hence, international organizations and programs urgently call for new research and methods to better understand the impacts of environment on child health and development. Substantially motivated by analytic needs arising from this research, the goal of this project is to develop a set of novel statistical models and methods and efficient algorithms to evaluate, interpret, and predict impacts of early life exposures to environmental toxins on child health and developmental outcomes. This new toolbox is general and applicable to analyze developmental mechanisms in studies of other health and clinical outcomes.

This proposal is motivated by the P.I.'s collaborative projects in environmental health sciences. One project aims to elucidate the complex interactions among exposures to mixtures of toxicants including endocrine disrupting chemicals (EDCs) as well as heavy metals and their effects on physical growth, sexual maturation and risk of metabolic syndrome. The other project aims to examine impacts of pre- and/or postnatal iron deficiency (ID) and other exposures (lead and pesticides) on children neurodevelopmental outcomes.

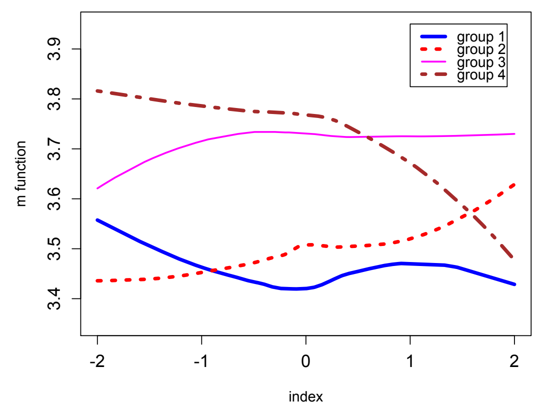
The existing statistical methodologies are inadequate to address several major analytic challenges that complicate the data analysis in the above projects. These complications include: (i) subjects are measured with simultaneous exposure to many (low-effect) toxicant agents, and (ii) subjects are recruited via four heterogeneous longitudinal cohorts over 18 years. The developed toolbox will allow us to assess effects of toxicant sets, instead of those of individual toxicants, to validate longitudinal cohorts in a high-dimensional data setting. These improvements in data analysis methodologies matter because new statistical models and more powerful methods can help to transform discovered signals into advances of understanding on children developmental mechanisms and behaviors.

Hence, in this proposal, the P.I. aims to develop a new class of semiparametric models (varying index coefficient models) to address several methodological needs, including (i) to determine if early life exposure to mixtures of toxicants alters child growth and development, (ii) if so, to identify which toxic agents, and in which functional forms, are responsible for the alteration, and (iii) to extend the proposed model to longitudinal outcomes and high-dimensional exposures. The P.I. has a wealth of experience and extensive publications on the development of semiparametric models and theory and methods in longitudinal and high-dimensional data analysis. The P.I. refers to her curriculum vitae for her experience and publications.

2 Methodology

2.1 Proposed approach

Regression analysis has a long history of utilizing interactions to examine altered covariate effects on outcomes, and our proposed approach differs in several ways from this classical approach, as detailed in the following discussion. Figure 2.1 displays four fitted curves against a linear combination of ten EDCs measured on mothers during the third trimester of pregnancy. They are obtained by fitting the existing single-index model (SIM): $E(Y|\mathbf{Z}) = m(\mathbf{Z}^T\boldsymbol{\beta})$ to each of four groups: (1) age$\leq$ 10, gender=boy; (2) age$\leq$ 10, gender=girl; (3) age$>$ 10, gender=boy; and (4) age$>$ 10, gender=girl, where Y is the logarithm of child weight, $\mathbf{Z}$ is a vector of the ten log-transformed EDCs. At around 10 yr of age, peak weight velocity in girls typically precedes onset of puberty. Clearly these estimated curves are not parallel and exhibit quite different patterns; for example, girl's weight appears to be more affected by mother's exposures. Compared to the other groups, girls aged 11-14 have much lower weight at higher level of the EDC index. Such preliminary evidence suggests the existence of nonlinear interaction effects of the EDC exposure with both age and gender. To analyze such features of scientific importance, we propose a class of new regression models allowing for nonlinear interactions, termed as *varying index coefficient model* (VICM), and establish a unified framework for joint estimation and inference for coefficient functions $m(\cdot)$ and loading coefficients $\boldsymbol{\beta}$.



For ease of exposition, we begin our discussion with cross-sectional data, and then extend the framework to longitudinal data. Let $\mathbf{Z} = (Z_1, \ldots, Z_p)^T$ be a vector of p toxicants and $\mathbf{X} = (X_1, \ldots, X_d)^T$ be a vector of d covariates with $X_1 = 1$. The proposed VICM takes the form:

$$Y = \sum\nolimits_{l=1}^{d} m_l\left(\mathbf{Z}_l^{\mathrm{T}}\boldsymbol{\beta}_l\right) X_l + \varepsilon, \text{ with } E\left(\varepsilon\, \mathbf{X}, \mathbf{Z}|\right) = 0, Var\left(\varepsilon\, \mathbf{X}, \mathbf{Z}|\right) = \sigma^2\left(\mathbf{X}, \mathbf{Z}\right),$$

where $\boldsymbol{\beta}_l = \left(\beta_{l1}, \ldots, \beta_{lp}\right)^T$ is a vector of unknown loading coefficients and $m_l(\cdot)$ are unknown smooth functions. In this model, all $\boldsymbol{\beta}_l$ coefficients will be calibrated with the response variable, fundamentally different from the currently popular means of forming an index (i.e. $\mathbf{Z}^{\mathrm{T}}\boldsymbol{\beta}$) via the principal component analysis (PCA). A weakness of PCA is that loading coefficients are determined with no use of the response variable, and consequently it is impossible to infer the association of individual toxicants with growth outcomes.

Task 1. Develop a generalized likelihood ratio (GLR) test for the hypothesis H_0 : $m_l(\cdot) = m_{\boldsymbol{\theta},l}(\cdot)$ versus H_a : $m_l(\cdot) \neq m_{\boldsymbol{\theta},l}(\cdot)$, where $m_{\boldsymbol{\theta},l}(\cdot)$ is a pre-specified function to reflect a form by which mixtures of toxicants affect the outcomes. For example, the case of $m_{\boldsymbol{\theta},l}(\cdot) = 0$ corresponds to X_l being an unimportant predictor and thus the $\boldsymbol{\beta}_l$ in the l-th index is no longer relevant. The GLR test statistic is constructed as

$$n\left\{\mathrm{RRS}\left(H_0\right) - \mathrm{RRS}\left(H_1\right)\right\} / \left\{2\mathrm{RRS}\left(H_1\right)\right\},$$

where RRS(H) is the residual sum of squares under the null or alternative hypothesis. We will establish the respective asymptotic distributions of the GLR statistic in two cases of the null hypotheses, namely constant and linear $m_{\boldsymbol{\theta},l}(\cdot)$ functions.

Task 2. Given that a function $m_l(\cdot)$ is not zero, to detect which toxicant agents are responsible for the alteration by testing a subset of $\boldsymbol{\beta}_l$ coefficients via $H_0 : \beta_{k_1 l} = \beta_{k_2 l} = \cdots = \beta_{k_K l} = 0$.

Tasks 1-2 will be built on the B-spline approximation and the profile least squares (PLS) approach. The PLS method allows us to avoid the back-ftting algorithm to handle the estimation of multiple nonparametric functions. In addition, the PLS enjoys the ease of developing rigorous large-sample properties as well as fast and stable numerical calculations.

Task 3. Construct confidence interval estimation for nonzero $m_l(\cdot)$ functions. The above one-step spline estimator based on the PLSE has no asymptotic distributions. To overcome, the P.I. proposes to develop a two-step spline backfitted local linear (SBLL) estimation by updating the one-step estimator through local kernel smoothing. The P.I. will establish both uniform consistency and asymptotic normality for the proposed SBLL estimator.

Task 4. Evaluate the utility of Bayes Information Criterion (BIC) to determine the number of interior knots in the PLSE and the utility of approximate mean integrated squared error (AMISE) to the bandwidth selection in the SBLL estimation.

2.2 Algorithm

The most critical computing work pertains to the optimization involved in the PLSE. We use the constrOptim.nl package in R software. Since this R package allows to supply user-specified gradients of the objective function, we derive explicit formulas to calculate all gradient terms. This can speed up computation significantly based on our preliminary simulation experience. To start the search, we use initial values obtained by using linear coefficient functions, which has been widely applied in the SIM. This algorithm development will be wrapped in a user-friendly R package.

2.3 Extension to Longitudinal Outcomes

The motivating projects collect longitudinal data (e.g. body composition in ELEMENT), $\left\{Y_i(t), \mathbf{X}_i(t) = \left(X_{i1}(t), \ldots, X_{id}(t)\right)^T, \mathbf{Z}_i(t) = \left(Z_{i1}(t), \ldots, Z_{ip}(t)\right)^T\right\}$, which are measured at time $t = t_{ij}$ ((the j-th visit for subject i), $j = 1, \ldots, m_i$, $i = 1, \ldots, n$. A longitudinal VICM is specified by the following semiparametric linear mixed-effects model (LMM):

$$Y_i(t_{ij}) = \sum\nolimits_{l=1}^{p} m_l\left(\mathbf{Z}_i(t_{ij})^{\mathrm{T}} \boldsymbol{\beta}_l\right) X_{il}(t_{ij}) + \mathbf{W}_i(t_{ij})^{\mathrm{T}} \mathbf{b}_i + \varepsilon_i(t_{ij}),$$

where $\mathbf{W}_i$ may be a subset of $\mathbf{X}_i$, and random effects $\mathbf{b}_i$ and error terms $\varepsilon_i(t_{ij})$ are assumed to be i.i.d. normally distributed with zero means and covariances $\mathbf{D}$ and $\sigma^2\mathbf{R}$, respectively. The standard methodology for estimation and inference in the LMM includes the maximum likelihood estimation (MLE) for the parameters in the mean model and the restricted maximum likelihood (REML) estimation for variance component parameters. A key methodological extension required by the longitudinal VICM is to generalize the classical MLE/REML method to a pro le MLE/REML under the B-spline approximation discussed above.

2.4 Extension to High-dimensional Exposures

Another extension is to develop a regularization method to handle high-dimensional toxicants in the above proposed models. One of the P.I.'s collaborative project measures more than 200 pesticides from blood samples, and it is believed that only a small proportion of the toxicants may be important in modifying regression coefficients. Inspired by the group LASSO method, the P.I. propose to regularize the PLSE or profile MLE/REML by the following penalty function: $\zeta_1 \sum_{l=1}^{d} \sqrt{\beta_{l1}^2 + \ldots + \beta_{lp}^2} + \zeta_2 \sum_{l=1}^{d} \sum_{k=1}^{p} |\beta_{lk}|$, where ζ_1 and ζ_2 are two tuning parameters. The first penalty term can effectively remove interactions with unimportant mixtures. The second penalty can identify important toxicants within the index. The regularization procedure will be based on the PLSE or profile MLE/REML with some necessary modifications on that.

3 Impact of the Hellman funds

The Hellman funds would enable the P.I. to (1) provide financial support to her and her graduate students to work on this project; (2) disseminate the research results through attending conferences, seminars and workshops, and publishing papers in peer-reviewed journals; (3) purchase access to datasets, computational resources and high performance computing; and (4) meet the P.I.'s collaborators and peers to discuss results and learn about cutting-edge research in the P.I.'s field. Thus, the P.I. believes that the Hellman funds would support the P.I. to a great extent to accomplish this project and it would be essential to her career. The proposed statistical methodology is novel and application-driven. The Hellman funds would help the P.I. continue working in this line of work and conducting intuitively appealing, theoretically meaningful and computationally efficient research.